

JEE Main - 4 | JEE - 2023 (Gen 1 & 2)

Date: 14/09/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 9. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

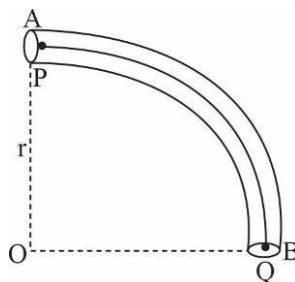
PART - I : PHYSICS

100 MARKS

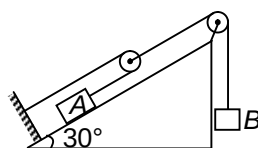
SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

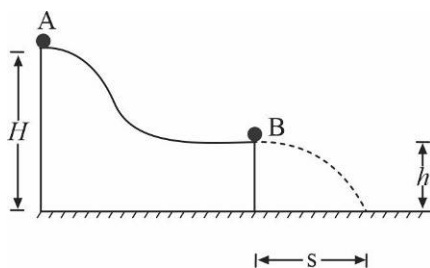
1. A smooth narrow tube in the form of an arc AB of circle of centre O and radius r is fixed so that A is vertically above O and OB is horizontal. Particles P of mass m and Q of mass $2m$ with a light inextensible string of length $(\pi r / 2)$ connecting them are placed inside the tube with P at A and Q at B and they are released from rest. Assuming the string remains taut during motion, the speed of particles when P reaches B, will be :



- (A) $\sqrt{\frac{2}{3}(1+\pi)gr}$ (B) $\sqrt{\frac{1}{3}(1+\pi)gr}$ (C) $\sqrt{(1+2\pi)gr}$ (D) $\sqrt{\frac{1}{2}(1+2\pi)gr}$
2. In the system shown in figure $m_B = 4$ kg and $m_A = 2$ kg. The pulleys are massless and friction is absent everywhere. The acceleration of block A is $\{g = 10 \text{ m/s}^2\}$:

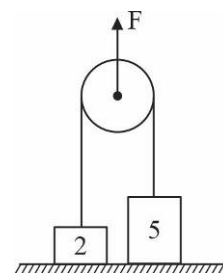


- (A) $\frac{10}{3} \text{ m/s}^2$ (B) $\frac{20}{3} \text{ m/s}^2$ (C) $\frac{5}{2} \text{ m/s}^2$ (D) $\frac{5}{3} \text{ m/s}^2$
3. An engine pumps water continuously through a hole. Speed with which water passes through the hole nozzle is v , and λ is the mass per unit length of the water jet as it leaves the nozzle. Find the rate at which kinetic energy is being imparted to the water.
- (A) $\frac{1}{2}\lambda v^2$ (B) $\frac{1}{2}\lambda v^3$ (C) $\frac{v^2}{2\lambda}$ (D) $\frac{v^3}{2\lambda}$
4. A small disc A slides down with initial velocity equal to zero from the top of a smooth hill of height H having a horizontal portion. What must be the height of the horizontal portion h to ensure the maximum distance s covered by the disc?



- (A) $h = \frac{H}{3}$ (B) $h = \frac{H}{2}$ (C) $h = \frac{H}{4}$ (D) $h = \frac{H}{6}$

5. Two blocks of masses 2 kg and 5 kg are at rest on ground. The masses are connected by a string passing over a frictionless pulley which is under the influence of a constant upward force $F = 50$ N. The accelerations of 5 kg and 2 kg masses are: $\{g = 10 \text{ m/s}^2\}$

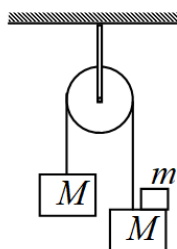


- (A) 0, 2.5 ms^{-2} (B) 0, 0
(C) 2.5 ms^{-2} , 2.5 ms^{-2} (D) 1 ms^{-2} , 2.5 ms^{-2}

6. If the potential energy between two molecules is given by $U = \frac{-A}{r^7} + \frac{B}{r^{14}}$, then at equilibrium, separation between molecules, is : $\{r$ is the distance between molecules $\}$

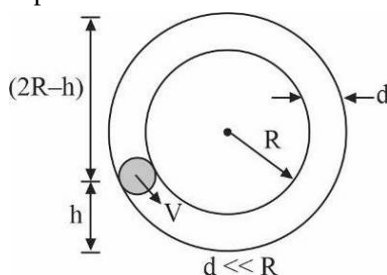
- (A) $(A/B)^{1/7}$ (B) $(2A/B)^7$ (C) $(2A/B)^{1/7}$ (D) $(2B/A)^{1/7}$

7. Two identical blocks of masses M are linked by a thread wrapped around a frictionless pulley as shown in the figure. A small block of mass m is placed on one of the blocks as shown. If pulley and strings are ideal, the force exerted by small block m on block M is:



- (A) $\frac{mMg}{2M+m}$ (B) $\frac{2mMg}{2M+m}$ (C) $\frac{2mMg}{M+m}$ (D) $\frac{2m^2g}{2M+m}$

8. With what minimum speed v must a small ball should be pushed inside a smooth vertical tube from a height h so that it may reach the top of the tube? Radius of the tube is R .



- (A) $v = \sqrt{3g(R-h)}$ (B) $v = \sqrt{2g(3R-2h)}$
(C) $v = \sqrt{3g(2R-2h)}$ (D) $v = \sqrt{2g(2R-h)}$

9. A body is displaced from origin to $(2m, 4m)$ under the influence of force $\vec{F} = (2x\hat{i} + 3y^2\hat{j}) \text{ N}$. Work done by the force will be: $\{x$ and y are coordinates in x and y direction where the particle is present $\}$

- (A) 20 J (B) 72 J (C) 68 J (D) 92 J

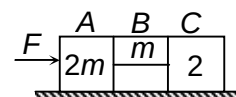
10. A uniform chain of length $4m$ is kept on a table such that a length of $1.2m$ hangs freely from the edge of the table. The total mass of the chain is 8 kg. What is the work done in pulling the entire chain on the table? $\{g = 10 \text{ m/s}^2\}$

- (A) 24 J (B) 3.6 J (C) 7.2 J (D) 14.4 J

11. An object of mass m has a speed v as it passes through the origin on its way out along with positive x -axis. The object is subjected to a retarding force given by $F = -Ax$ ($A > 0$) (x is the x co-ordinate of the object). The x co-ordinate of the object when it stops is:

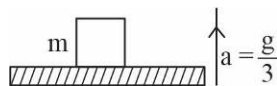
(A) $v\sqrt{\frac{m}{A}}$ (B) $v\sqrt{\frac{2m}{A}}$ (C) $v\sqrt{\frac{m}{2A}}$ (D) $2v\sqrt{\frac{m}{A}}$

12. The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C . Friction coefficient between B and C is μ . Minimum value of F to prevent block B from slipping is:



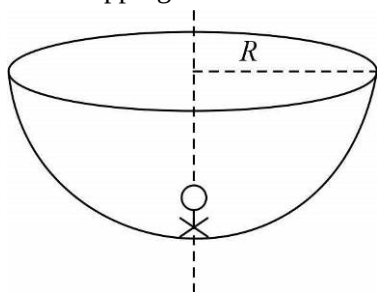
(A) $\left(\frac{3}{2\mu}\right)mg$ (B) $\left(\frac{5}{2\mu}\right)mg$ (C) $\left(\frac{5}{2}\right)\mu mg$ (D) $\left(\frac{3}{2}\right)\mu mg$

13. A block of mass m is kept on a platform which starts from rest with constant acceleration $g/3$ upwards, as shown in fig. Work done by normal reaction on block in time t is:



(A) $\frac{5mg^2t^2}{18}$ (B) $-\frac{mg^2t^2}{3}$ (C) $\frac{2mg^2t^2}{9}$ (D) Zero

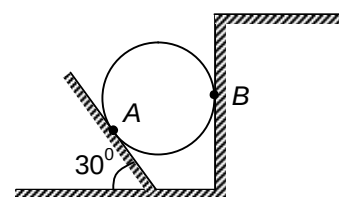
14. An insect is at the centre of the hemispherical bowl as shown in the figure. The mass of the insect is m and the coefficient of friction between insect and bowl is μ . The insect starts crawling to the top of the bowl. Find the displacement between the points where the insect is standing now to the point where it starts slipping.



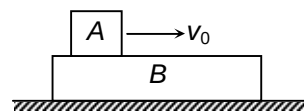
(A) $\sqrt{2}R\left(\frac{\sqrt{1+\mu^2}-1}{\sqrt{1+\mu^2}}\right)^{1/2}$ (B) $\frac{R\mu}{\sqrt{1+\mu^2}}$
(C) $R - \frac{R}{\sqrt{1+\mu^2}}$ (D) None of these

15. The 50 kg homogeneous smooth sphere rests on the 30° incline A and bears against the smooth vertical wall B. The contact force at B is ($g = 10 \text{ m/s}^2$):

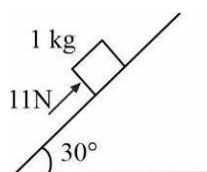
(A) 250 N (B) zero
(C) $\frac{500}{\sqrt{3}}$ N (D) 500 N



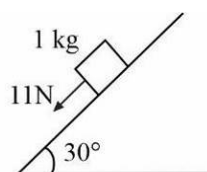
16. A block A of mass m is placed over a plank B of mass $2m$. Plank B is placed over a smooth horizontal surface. The coefficient of friction between A and B is $\frac{1}{2}$. Block A is given a velocity v_0 towards right. Acceleration of B relative to A is:



- (A) $\frac{g}{2}$ (B) g (C) $\frac{3g}{4}$ (D) zero
17. A balloon of weight W newton descends with an acceleration $f \text{ m/s}^2$. The weight that must be thrown out in order to give equal upward acceleration to the balloon will be:
- (A) $\frac{Wf}{g}$ (B) $\frac{2Wf}{g}$ (C) $\frac{2Wf}{g+f}$ (D) $\frac{W(g+f)}{f}$
18. A mass m moves with a velocity v and collides with another identical mass. After collision, the first mass moves with velocity $v/2$ in a direction perpendicular to the initial direction of motion. Find the speed of the second mass after collision.
- (A) $2v$ (B) $\frac{v}{2}$ (C) $\frac{\sqrt{5}}{2}v$ (D) $(\sqrt{2}+1)v$
19. A block of mass 1 kg is kept on a rough inclined plane as shown in the figure. A person can apply maximum force of 11 N up the inclined as shown in figure 1 to keep it at rest. What is the acceleration of the same block if same 11 N force is applied downwards as shown in figure 2? $\{g=10 \text{ m/s}^2\}$ $\{\mu_s = \mu_k\}$



(Figure 1)



(Figure 2)

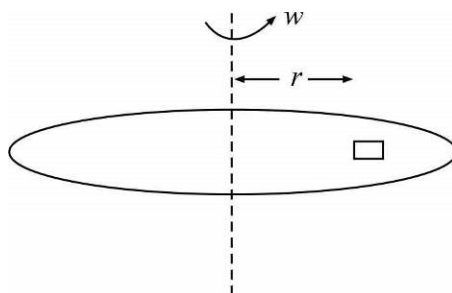
- (A) 16 m/s^2 (B) 11 m/s^2 (C) 10 m/s^2 (D) 5 m/s^2
20. A particle is suspended from a fixed point by a string of length 5 m . It is projected from the equilibrium position with such a velocity that the string slackens after the particle has reached a height 8 m above the lowest point. The maximum height from the lowest point, the particle can reach will be : $(g=10 \text{ m/s}^2)$
- (A) $\left(8 + \frac{1}{25}\right)m$ (B) $\left(9 - \frac{1}{25}\right)m$ (C) $\left(9 + \frac{24}{25}\right)m$ (D) $\left(8 + \frac{2}{25}\right)m$

SPACE FOR ROUGH WORK

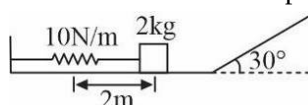
SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

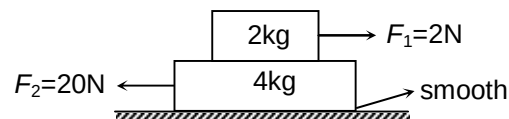
21. A particle of mass 20 kg is moving along x-axis according to the relation $x = 5t^3 + 3t + 4$. Impulse (in Newton-second) of the force acting on the particle over a time-interval between $t = 0$ and $t = 0.1$ second will be :
22. A coin is kept on a rough turntable rotating with angular speed ω . It is just about to slip on the turntable when it is at a distance r from the centre of turntable. If the turntable rotates with an angular speed $\frac{\omega}{3}$, then it will slip at a distance of nr . Find n .



23. In the figure shown, all surfaces are smooth and force constant of spring is 10 N/m. Block of mass 2 kg is not attached with the spring. The spring is compressed by 2m and then released. The maximum distance (in meter) travelled by the block over the inclined plane will be: (Take $g = 10 \text{ m/s}^2$)



24. In the arrangement shown in figure, coefficient of friction between the two blocks $\mu = \frac{1}{2}$. The force of friction (in Newtons) acting between the two blocks is: $\{g = 10 \text{ m/s}^2\}$

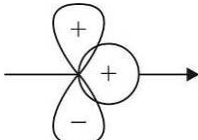
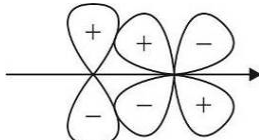
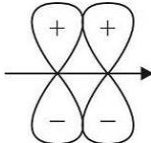
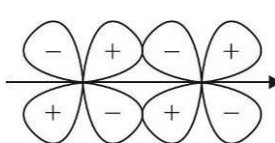


25. A particle of mass 1 kg is moving in one dimension under a force that delivers a constant power 6 W to the particle. If initial speed of the particle is zero, the speed (in ms^{-1}) after 3s is :

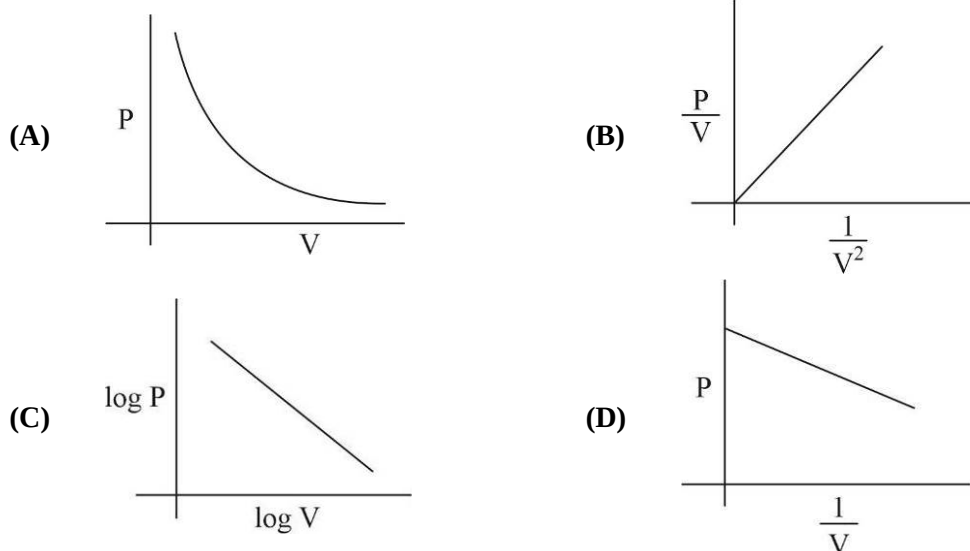
SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

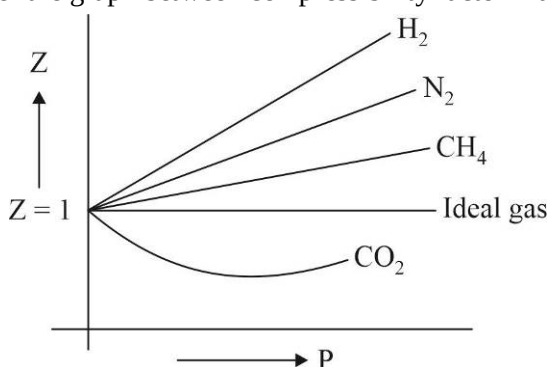
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- The polarity of (i) PF_3Cl_2 and (ii) PF_2Cl_3 respectively are ?
 (A) (i) Non Polar (ii) Polar (B) (i) Polar (ii) Non Polar
 (C) (i) Non Polar (ii) Non Polar (D) (i) Polar (ii) Polar
- In XeF_2 , XeF_4 , XeF_6 the number of lone pairs on Xe are respectively :
 (A) 2, 3, 1 (B) 1, 2, 3 (C) 4, 1, 2 (D) 3, 2, 1
- Which of the following is not isostructural with SiCl_4 ?
 (A) SO_4^{2-} (B) PO_4^{3-} (C) NH_4^+ (D) SCl_4
- The $p\pi - p\pi$ bond is present in :
 (A) XeO_3 (B) SO_4^{2-} (C) SO_2 (D) All of these
- Which of the following is a positive overlap that leads to bonding ?
 I.  II. 
 III.  IV. 
 (A) I and II (B) II and III (C) III and IV (D) I and IV
- The species that cannot exist is :
 (A) SiF_6^{2-} (B) BF_6^{3-} (C) SF_6 (D) AlF_6^{3-}
- According to the molecular orbital theory which of the following statement is incorrect ?
 [LUMO = lowest unoccupied molecular orbital]
 (A) LUMO level for C_2 molecule is $\sigma 2p_z$ orbital
 (B) In C_2 molecules both the bonds are π bonds
 (C) In C_2^{2-} ion there is one σ and two π bonds
 (D) C_2 is paramagnetic but C_2^{2-} is diamagnetic
- According to the kinetic theory of gases, between two successive collisions, a gas molecules travels :
 (A) In a circular path (B) In a zig-zag path
 (C) In a straight line path (D) With negligible speed
- Ratio of root mean square speed and average speed of an ideal gas is :
 (A) $\left[\frac{3\pi}{4}\right]^{1/2}$ (B) $\left[\frac{3\pi}{8}\right]^{1/2}$ (C) $\left[\frac{5\pi}{8}\right]^{1/2}$ (D) $\left[\frac{\pi}{3}\right]^{1/2}$

10. The rates of diffusion of gases A and B of molecular weight 36 and 64 are in the ratio :
 (A) 9 : 16 (B) 4 : 3 (C) 3 : 4 (D) 16 : 9
11. Which of the following graph does not represent Boyles law ?



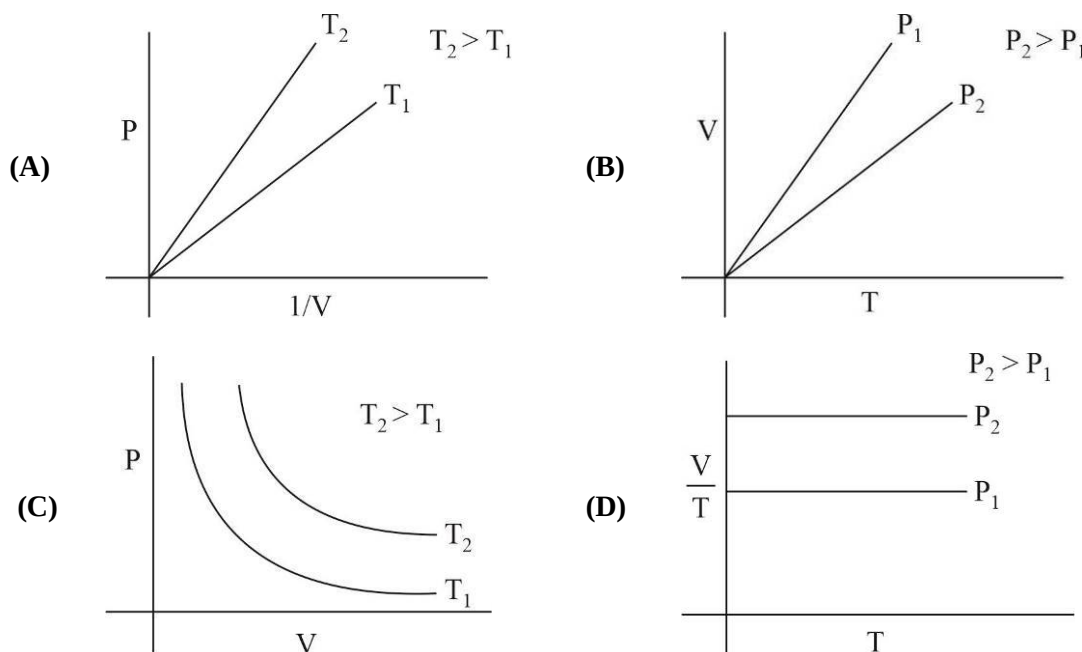
12. Which of the following statement is true at very high pressure for a real gas ?
 (A) Compressibility factor is greater than 1
 (B) Molar volume occupied by the gas is more as compared to ideal gas at similar pressure and temperature
 (C) The gas is less compressible as compared to an ideal gas
 (D) All of these
13. Equal moles of two gases A and B were enclosed in a vessel at certain temperature T . The rate of effusion of the gases through a small orifice is 1 : 4. Another sample of the gases, having A and B in 2 : 3 molar ratio is kept in another vessel and allowed to effuse at same temperature T . The molar ratio of these gases which is coming out in the initial effusing mixture is :
 (A) 1 : 6 (B) 1 : 12 (C) 1 : 18 (D) 1 : 24
14. Consider the graph between compressibility factor Z and pressure P :



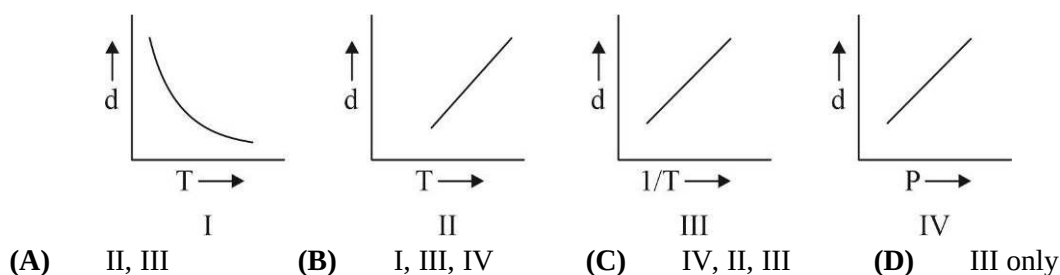
The correct increasing order of ease of liquefaction of the gases shown in the above graph is :

- (A) $H_2 < N_2 < CH_4 < CO_2$ (B) $CO_2 < CH_4 < N_2 < H_2$
 (C) $H_2 < CH_4 < N_2 < CO_2$ (D) $CH_4 < H_2 < N_2 < CO_2$

15. Which of the following graphs is inconsistent for fixed quantity of an ideal gas ?



16. Which of the following graphs is correct for an ideal gas ?



17. Species with formula IF_x^n is planar and non-polar, then possible values of x and n are :
(Where x is number of F atoms and n is charge on species)

- (A) $x = 2$ and $n = +1$ (B) $x = 3$ and $n = 0$
(C) $x = 2$ and $n = -1$ (D) $x = 5$ and $n = 0$

18. The correct order of 'S–O' bond length is :

- (A) $SO_3^{2-} < SO_4^{2-} < SO_3$ (B) $SO_3^{2-} > SO_4^{2-} > SO_3$
(C) $SO_4^{2-} > SO_3^{2-} > SO_3$ (D) $SO_4^{2-} > SO_3 > SO_3^{2-}$

19. At 300K and 1atm, 10 ml of hydrocarbon required 75 ml of O_2 for complete combustion and 50 ml of CO_2 is formed. The formula of hydrocarbon is :

- (A) C_4H_6 (B) C_5H_{10} (C) C_5H_8 (D) C_4H_8

20. Which of the following has highest bond angle between any two bonds containing the central N-atom ?

- (A) NH_3 (B) NE_3 (C) $N(CH_3)_3$ (D) $N(SiH_3)_3$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. What is the ratio of number of nodal planes present in π_{2p}^* and σ_{1s}^* molecular orbitals ?
22. Number of molecules among the following having non-zero dipole moment is :
 XeO_3 , SO_3 , SF_4 , SF_6 , H_2S , CS_2 , SO_2 , H_2O
23. 100 ml of $\text{O}_2(\text{g})$ effuse from a porous container in 100 sec. 75 ml of unknown gas effuses under the same conditions of temperature and pressure in 150 sec. Vapour density of unknown gas is $8x$. Value of x is :
24. A spherical balloon of radius 3cm containing helium gas has a pressure of 48×10^{-3} bar. At the same temperature, the pressure of a spherical balloon of radius 6cm containing the same amount of the gas will be _____ $\times 10^{-3}$ bar.
25. Among the given list of molecules, identify number of species in which the central atom is showing hybridisation in its excited electronic configuration:
 NH_4 CH_4 H_2O BF_3 PCl_5

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS

100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- In a $\triangle ABC$, side AB has the equation $2x + 3y = 29$ and the side AC has the equation $x + 2y = 16$. If the mid point of BC is (5, 6), then the equation of BC is :
 (A) $2x + y = 16$ (B) $x + y = 11$ (C) $2x - y = 4$ (D) $x + y = 10$
- The line parallel to the x-axis and passing through the intersection of the lines $ax + 2by + 3c = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is :
 (A) above the x-axis at a distance of $\left(\frac{2}{3}\right)$ from it
 (B) above the x-axis at a distance of $\left(\frac{3}{2}\right)$ from it
 (C) below the x-axis at a distance of $\left(\frac{2}{3}\right)$ from it
 (D) below the x-axis at a distance of $\left(\frac{3}{2}\right)$ from it
- In how many ways can 5 boys and 3 girls sit in a row so that no two girls are together ?
 (A) $5! \times 3!$ (B) ${}^4P_3 \times 5!$ (C) ${}^6P_3 \times 5!$ (D) ${}^5P_3 \times 3!$
- The number of words which can be formed using all the letters of the word LUCKNOW when the vowels always occupy even places is :
 (A) 120 (B) 720 (C) 400 (D) 420
- A library has a copies of one book, b copies each of two books, c copies each of three books, and single copy of d books. The total number of ways in which these books can be arranged in a shelf, is equal to :
 (A) $\frac{(a + 2b + 3c + d)!}{a!(b!)^2(c!)^3}$ (B) $\frac{(a + 2b + 3c + d)!}{a!(2b!)(c!)^3}$
 (C) $\frac{(a + b + 3c + d)!}{(c!)^3}$ (D) $\frac{(a + 2b + 3c + d)!}{a!(2b)!(3c)!}$
- There are 3 candidates for a post and one is to be selected by the votes of 7 men. The number of ways in which votes can be given is : [A man has to vote for any one candidate]
 (A) 7^3 (B) 3^7 (C) 7C_3 (D) 7P_3
- For a game in which two partners oppose two other partners, eight men are available. If every possible pair must play with every other pair, the number of games played is :
 (A) ${}^8C_2 \cdot {}^6C_2$ (B) ${}^8C_2 \cdot {}^6C_2 \cdot 2$ (C) ${}^8C_4 \cdot 3$ (D) ${}^8C_2 \cdot 3$
- There are 12 volleyball players in all in a college, out of which a team of 9 players is to be formed. If the captain always remains the same, then in how many ways can the team be formed ?
 (A) 36 (B) 108 (C) 99 (D) 165

9. If m_1, m_2 be the roots of the equation $x^2 + (\sqrt{3} + 2)x + \sqrt{3} - 1 = 0$, then the area of the triangle formed by the lines $y = m_1x, y = m_2x$ and $y = 2$ is :
- (A) $\sqrt{33} - \sqrt{11}$ sq. units (B) $\sqrt{11} + \sqrt{33}$ sq. units
(C) $2\sqrt{33}$ sq. units (D) 121 sq. units
10. If the lines $x + y + 1 = 0$; $4x + 3y + 4 = 0$ and $x + \alpha y + \beta = 0$, where $\alpha^2 + \beta^2 = 2$, are concurrent then:
- (A) $\alpha = 1, \beta = -1$ (B) $\alpha = 1, \beta = \pm 1$ (C) $\alpha = -1, \beta = \pm 1$ (D) $\alpha = \pm 1, \beta = 1$
11. Let a, b, c and d be non-zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes, then :
- (A) $2bc - 3ad = 0$ (B) $2bc + 3ad = 0$ (C) $2ad - 3bc = 0$ (D) $3bc + 2ad = 0$
12. The set of real values of k for which the lines $x + 3y + 1 = 0, kx + 2y - 2 = 0$ and $2x - y + 3 = 0$ form a triangle is :
- (A) $R - \left\{-4, \frac{2}{3}\right\}$ (B) $R - \left\{-4, \frac{-6}{5}, \frac{2}{3}\right\}$
(C) $R - \left\{\frac{-2}{3}, 4\right\}$ (D) R
13. The centroid of a triangle lies at the origin and the coordinates of its two vertices are $(-8, 7)$ and $(9, 4)$. The area of the triangle is :
- (A) $\frac{95}{6}$ (B) $\frac{285}{2}$ (C) $\frac{190}{3}$ (D) 285
14. The number of ways of arranging m positive and $n < m + 1$ negative signs in a row so that no two negative signs are together is :
- (A) ${}^{m+1}P_n$ (B) ${}^{n+1}P_m$ (C) ${}^{m+1}C_n$ (D) ${}^{n+1}C_m$
15. The total number of ways of selecting 10 balls out of an unlimited number of identical white, red and blue balls is equal to :
- (A) ${}^{12}C_2$ (B) ${}^{12}C_3$ (C) ${}^{10}C_2$ (D) ${}^{10}C_3$
16. The number of 3 letter words which can be formed with two different consonants and one vowel out of 7 different consonants and 3 different vowels, the vowel should be between the two consonants is :
- (A) 92 (B) 172 (C) 126 (D) 136
17. The number of non-negative integral solutions to the system of equation $x + y + z + u + t = 20$ and $x + y + z = 5$ is:
- (A) 336 (B) 346 (C) 246 (D) None of these
18. Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If origin is the orthocenter of the triangle, then the third vertex is:
- (A) $(4, 7)$ (B) $(-4, 7)$ (C) $(4, -7)$ (D) $(-4, -7)$
19. P is a point on the line $y + 2x = 1$ and Q and R are two points on the line $3y + 6x = 6$ such that the triangle PQR is an equilateral triangle. Area of this triangle is equal to:
- (A) $\frac{1}{15}$ (B) $\frac{1}{5\sqrt{3}}$ (C) $\frac{1}{3\sqrt{5}}$ (D) $\frac{1}{2\sqrt{15}}$

20. Given the family of lines, $a(2x + y + 4) + b(x - 2y - 3) = 0$. Among the lines of the family, the number of lines situated at a distance of $\sqrt{10}$ from the point $M(2, -3)$ is:
- (A) 0 (B) 1 (C) 2 (D) ∞
-

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 9 (both inclusive).

21. How many words can be made by using all letters of word 'BAHUBALI' if all words start and end with vowels, then sum of digits in the number is _____.
22. If ${}^nC_3 + {}^nC_4 > {}^{n+1}C_3$, then least integral value of n is _____.
23. The number of 4-digit numbers that can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7 so that each number contains digit 1 is N , then unit place of N is _____. (Repetition of digits not allowed)
24. The point $(a^2, a+1)$ lies in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin if, then number of integral value of a is _____.
25. If the three families of lines
- $$(y - 2x + 1) + \lambda_1(2y - x - 1) = 0$$
- $$(3y - x - 6) + \lambda_2(y - 3x + 6) = 0$$
- and $(ax + y - 2) + \lambda_3(6x + ay - a) = 0 \quad (a > 0)$
- have a common line, then the value of a is:
-

SPACE FOR ROUGH WORK

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